

GAYATRI VIDYA PARISHAD
COLLEGE FOR DEGREE AND PG COURSES(A)
 (Affiliated to Andhra University), VISAKHAPATNAM
ENGINEERING & TECHNOLOGY PROGRAM
I/IV B. Tech. I Semester Regular Examinations, SEP-2023
Common for CE / CSE /CSE(AIML) / ECE / ME
SUBJECT :: ENGINEERING MATHEMATICS – II (22092111)
(W.e.f 2022-23 admitted batch)

Date: 04-09-2023(A.N) Time: 02:00pm to 05:00pm Max. Marks: 70

PART - A

Answer ALL the questions

5 × 2 = 10M

- 1 a) Define rank of a Matrix. [2M]
- b) Find the Eigen values of a matrix A where $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ [2M]
- c) The integrating factor of the differential equation $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$. [2M]
- d) The particular integral of the differential equation $(D^2 + 5D + 6)y = e^x$. [2M]
- e) Find Laplace transform of $\sin 2t \sin 3t$ [2M]

PART - B

Answer ALL the questions

5 × 12 = 60 M

- 2(a) Find the rank of a matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ by converting it into Echelon form. [6M]

- (b) Solve the system of $x + y + z = 4, x - y + z = 0, 2x + y + z = 5$ by using Gauss elimination method. [6M]

OR

3(a)

[6M]

Find the rank of a matrix $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ by converting it into Normal form.

(b) Solve the system of

[6M]

$2x - y + 3z = 8$, $x - 2y - z = -4$, $3x + y - 4z = 0$ by using LU factorization method.

4(a) Find the Eigen values and the corresponding Eigen

[6M]

vectors of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

(b) Using Cayley – Hamilton theorem, find A^4 where

[6M]

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

OR

5 Reduce the quadratic form $8x^2 + 7y^2 + 3z^2 - 12xy + 4xz - 8yz$ to canonical form by an orthogonal transformation and give the matrix of transformation. Also, state the nature of the quadratic form

[12 M]

6(a) Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$

[6M]

(b) If the air is maintained at 30°C and the temperature of the body cools from 80°C to 60°C in 12 minutes, find the temperature of the body after 24 minutes.

[6M]

OR7(a) Solve $x \frac{dy}{dx} + y = x^3 y^6$

[6M]

(b) Find the orthogonal trajectories of the family of Coaxile circles $x^2 + y^2 + 2\lambda x + c = 2$, λ being the parameter.

[6M]

8(a) Solve $(D-2)^2 y = 8e^{2x} + 8\sin 2x$ [6M]

(b) Solve $\frac{d^2 y}{dx^2} + 4y = \tan 2x$ by using the method of variation of parameters. [6M]

OR

9(a) Solve $\frac{d^2 y}{dx^2} - 4y = x \sinh x$ [6M]

(b) Solve the Legendere's linear equation [6M]

$$(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 2 \sin[\log(1+x)]$$

10(a) Find the Laplace Transform to the following functions [6M]

(i) $t^2 \sin at$ (ii) $te^{-t} \sin 3t$

(b) Find the Laplace Transform to the following functions [6M]

$$\int_0^{\infty} e^{-t} \left(\frac{\cos at - \cos bt}{t} \right) dt \quad \text{(ii)} \quad e^{-t} \int_0^t \frac{\sin t}{t} dt$$

OR

11(a) Find the inverse Laplace Transform of [6M]

$$\frac{s}{(s^2 + 1)(s^2 + 4)}$$

(b) Solve $y''' + 2y'' - y' - 2y = 0$ given [6M]
 $y(0) = y'(0) = 0$ and $y''(0) = 6$ by the method of Laplace Transforms

[R-22]

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ENGINEERING & TECHNOLOGY PROGRAM

I/IV B.Tech Semester – II Supplementary Examinations, Nov-2025

Common For CE,CSE,CSE(AI&ML),ECE and ME

SUBJECT :: ENGINEERING MATHEMATICS-II (22092111)

(W.e.f 2022-2023 admitted batch)

Date:05-11-2025 (A.N) Time:02:00 PM to 05:00 PM Max. Marks: 70

PART - A

1 Answer ALL the questions **5 x 2 = 10M**

- a) Define Echelon form of a matrix. [2M]
- b) Find the Eigen values of the matrix $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. [2M]
- c) Find the orthogonal trajectories of the family of straight lines that are passing through the origin. [2M]
- d) Find the particular integral of $(D^2+1)y=100$. [2M]
- e) Find the Laplace transform of $e^{-2t}(\sin 3t+2\cos 5t)$. [2M]

PART - B

Answer ALL the questions

5 x 12= 60M

- 2a) Find the rank of a matrix $A = \begin{bmatrix} 2 & 1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$ by reducing it to normal form. [6M]

- b) Using Gauss-elimination method, solve the following system of equations $2x+y+z=10$, $3x+2y+3z=18$, $x+4y+9z=16$. [6M]

OR

- 3a) Solve the system of equations $10x+y+z=12$, $2x+10y+z=13$, $2x+2y+10z=14$ by LU Factorization method. [6M]

- b) Define rank and find the rank of the matrix A by reducing [6M]

it in to its echelon form where $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$.

- 4a) Find the inverse of the matrix, using Cayley-Hamilton [6M]

theorem $A = \begin{bmatrix} 8 & -8 & 2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$.

- b) Find the nature, rank and signature of the quadratic form [6M]
 $10x^2 + 2y^2 + 5z^2 - 4xy - 10zx + 6yz$.

OR

- 5a) Find the Eigen values and the corresponding Eigen vectors [6M]

of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ 6 & 7 & 4 \\ 2 & -4 & 3 \end{bmatrix}$.

- b) For the matrix $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$ find the Eigen values of [6M]
 $3A^3 + 5A^2 - 6A + 2I$.

- 6a) Solve the differential equation $\frac{dy}{dx} + y = e^{ex}$. [6M]

- b) The temperature of a body drops from 100°C to 75°C in 10 [6M]
minutes, when the surrounding air is at 20°C . Find the
temperature of the body after half an hour.

OR

- 7a) Solve the differential equation $x \frac{dy}{dx} + y = x^3 y^6$. [6M]

- b) Show that the family of curves $y^2 = 4a(x+a)$ are self- [6M]
orthogonal.

8a) Solve the differential equation $(D^2-3D-2)y=\cosh x$. [6M]

b) Solve by the method of variation of parameters [6M]
 $(D^2+1)y=\operatorname{cosec} x$.

OR

9a) Solve the differential equation $(x^2D^2+4xD+2)y=\sin(\log x)$. [6M]

b) Solve the differential equation $(D^2+2D+1)y=x \cos x$. [6M]

10a) Find the Laplace transform of $\frac{\cos 4t \sin 2t}{t}$. [6M]

b) Using Laplace transforms evaluate $\int_0^{\infty} t e^{-t} \sin t \, dt$. [6M]

OR

11a) Find the inverse Laplace transform of $\left[2+s \log\left(\frac{s-1}{s+1}\right)\right]$. [6M]

b) Solve $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+5y=5$ with the conditions $y(0)=0, y'(0)=0$. [6M]

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I/IV B.Tech Semester – II Supplementary Examinations, Feb-2024

Common For CE,CSE,CSE(AIML),ECE and ME

SUBJECT :: ENGINEERING MATHEMATICS- II (22092111)

(W.e.f 2022-2023 admitted batch)

Date:06-02-2024 (F.N) Time:10:00 AM to 01:00 PM Max. Marks: 70

PART - A

1 Answer ALL the questions **5 x 2 = 10M**

- a) Define Normal form of a matrix.
- b) Find the Eigen values of the matrix $\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$.
- c) Find the orthogonal trajectories of the family of parabolas $y^2=4ax$.
- d) Find the particular integral of $(D^3+4D)y=\sin 2x$.
- e) Find the Laplace transform of $t \sin^2 t$.

PART - B

Answer ALL the questions **5 x 12= 60M**

- 2a) Find the rank of a matrix $A = \begin{bmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ by reducing it

to normal form.

- b) Solve the following system of equations $2x+y+z=10$,
 $3x+2y+3z=18$, $x+4y+9z=16$.

OR

- 3a) Find the value of λ for which the system of equations
 $3x-y+4z=3$, $x+2y-3z=-2$, $6x+5y+\lambda z=-3$ will have infinite
number of solutions and solve them with that λ value.

b) Solve the system of equations $10x+y+z=12$, $2x+10y+z=13$, $2x+2y+10z=14$ by Gauss-elimination method.

4a) Compute the inverse of the matrix, using Cayley-Hamilton theorem, $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 2 \end{pmatrix}$.

b) Find the Eigen values and the corresponding Eigen vectors of the matrix, $A = \begin{pmatrix} 5 & -2 & 0 \\ -2 & 6 & 2 \\ 0 & 2 & 7 \end{pmatrix}$.

OR

5. Reduce the quadratic form $3x^2+5y^2+3z^2-2yz+2zx-2xy$ to the canonical form and specify the matrix of transformation.

6a) Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$.

b) The temperature of a body drops from 80°C to 60°C in 20 minutes, when the surrounding air is at 40°C . Find the temperature of the body after 40 minutes.

OR

7a) Solve the differential equation $x \frac{dy}{dx} + y = x^3 y^6$.

b) Find the orthogonal trajectories of the family of curves $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$.

8a) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x$.

b) Apply the method of variation of parameters to solve

$$\frac{d^2 y}{dx^2} + a^2 y = \tan ax.$$

OR

9a) Solve the differential equation $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - 3y = x^2 e^{-3x}$.

b) Solve the differential equation $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$.

10a) Find the Laplace transform of $t^2 e^{-t} \cos^2 t$.

b) Using Laplace transforms evaluate $\int_0^{\infty} \frac{e^{-at} - e^{-bt}}{t} dt$.

OR

11a) Find the inverse Laplace transform of $\frac{s^2}{(s-a)(s-b)}$.

b) Using Laplace transform, solve

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = e^t \sin t, \text{ with the conditions } y(0)=0 \text{ and } y'(0)=1.$$

[R-22]

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I/IV B.Tech Semester – II Supplementary Examinations, Dec-2024

Common For CE,CSE,CSE(AI&ML),ECE and ME

SUBJECT :: ENGINEERING MATHEMATICS – II (22092111)

(W.e.f 2022-2023 admitted batch)

Date:06-12-2024 (A.N) Time:02:00 PM to 05:00 PM Max. Marks: 70

PART - A

1 Answer ALL the questions **5 x 2 = 10M**

a) Define normal form of a matrix. [2M]

b) Find the Eigen values of a matrix A and A^2 where [2M]

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}.$$

c) The integrating factor of the differential equation [2M]

$$x \frac{dy}{dx} + y = x^3 y^6.$$

d) The solution of the differential equation $(D^2 + 5D + 6)x = 0$. [2M]

e) Find Laplace transform of $\sin^3 3t$. [2M]

PART - B

Answer ALL the questions **5 x 12 = 60M**

2a) Find the rank of a matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ by converting it [6M]

into normal form.

b) Solve the following system of equations [6M]
 $x + 3y + 6z = 2, 3x - y + 4z = 9, x - 4y + 2z = 7$ by using
LU factorization method.

OR

3a) Find the rank of a matrix $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$ by converting it into normal form. [6M]

b) Solve the following system of equations $2x - 3y + 4z = -4$, $x + z = -1$, $-y + 4z = 2$ by using Gauss elimination method. [6M]

4a) Find the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. [6M]

b) Using Cayley - Hamilton theorem, find A^{-1} where $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$. [6M]

OR

5) Reduce the quadratic form $3x^2 + 3y^2 + 3z^2 + 2xy + 2xz - 2yz$ to canonical form by an orthogonal transformation and give the matrix of transformation. Also, state the nature of the quadratic form. [12M]

6a) Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$. [6M]

b) If 30% of radioactive substance disappeared in 10 days, how long will it take for 90% of it to disappear. [6M]

OR

7a) Solve $xy(1 + xy^2)\frac{dy}{dx} = 1$. [6M]

b) Find the orthogonal trajectories of the family of semi-cubical parabolas $ay^2 = x^3$, a being the parameter. [6M]

8a) Solve $(D^3+1)y = \cos(2x-1)$. [6M]

b) Solve $(D^4+2D^2+1)y = x^2 \cos x$. [6M]

OR

9a) Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x \sin x$. [6M]

b) Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = \frac{e^{3x}}{x^2}$ by using the method of variation of parameters. [6M]

10a) Find the Laplace Transform to the following functions. [6M]

i) $t^2 \cos at$ ii) $\int_0^\infty te^{-3t} \sin t \, dt$

b) Find the Laplace Transform to the following function. [6M]

$\int_0^t \int_0^t \int_0^t (t \sin t) dt dt dt$.

OR

11a) Find the inverse Laplace Transform of $\log \frac{s^2+1}{s(s+1)}$. [6M]

b) Solve $y''' - 3y'' + 3y' - y = t^2 e^t$ given [6M]

$y(0)=1, y'(0)=0$ and $y''(0)=-2$ by the method of Laplace Transforms.
